Sanyam Sharma

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Work Experience

Indian Institute of Technology, Hyderabad — Hyderabad, Telangana
Research Intern — Jan 2026 – Present
- Working at NetX Lab @ IIT Hyderabad

Indian Institute of Technology, Mandi — Mandi, Himachal Pradesh
Summer Intern — Jun 2025 – Jul 2025
- Extracted features from audio data using MFCC and HF pretrained models (wav2vec2.0, wav2vec2-phoneme)
- Implemented Conformer-based deep learning model in PyTorch for spoken language identification
- Achieved up to 79% accuracy on unseen language identification data
- Developed a PyQt6 desktop GUI and integrated pretrained models for spoofed speech detection

Indian Institute of Technology, Mandi — Mandi, Himachal Pradesh
Research Intern — Jun 2024 – Jun 2024
- Set up data pipeline using Telegraf and InfluxDB to collect and store sensor data from an ESP32 via MQTT .
- Deployed the system on a Raspberry Pi using Docker for real-time data processing.

Education

Jawaharlal Nehru Government Engineering College — Mandi, Himachal Pradesh
B.Tech – CSE (AI & ML) — 2022 – 2026
Key Courses: DSA, Machine Learning, Deep Learning, Cloud Computing, Generative AI

Government Senior Secondary School, Kangoo — Hamirpur, Himachal Pradesh
Senior Secondary (Non-Medical) — 2022

Projects

AI-Powered Assistive Vision for Blind People - Major Project — Aug 2025 – Dec 2025
Tech: gTTS, FaceNet, Gemini-API, pytesseract, RasberryPi
- Developed an AI-powered assistive vision smart cap for blind person using Raspberry Pi, camera, LiDAR, audio module
- Implemented OCR-based text recognition to read printed and convert it to audio output in real time
- Integrated obstacle detection with audio-based navigation for safe mobility

Extension for malicious QR code and URL detection - Capstone Project — Jan 2025 – May 2025
Tech: Python, Flask, Scikit-Learn, JavaScript
- Extracted and engineered features from large scale malicious and legitimate URL/QR dataset
- Built a stacked ensemble ML model (5 base + meta-classifier), achieving 91.3% accuracy
- Developed Chrome extension with Flask backend for model inference

SpoofedSpeechGUI | GitHub
Tech: Python, PyQt6, PyTorch, Librosa
- Extracted features from the raw audio using Librosa
- Applied pre-trained AASIST and RawNet models for spoofed audio detection
- Developed a PyQt6 desktop GUI application

RNN-QA | GitHub | Docker
Tech: Docker, FastAPI, Python, PyTorch
- Trained a single layer RNN for question answer dataset
- Wrapped model in FastAPI for inferencing
- Published a Docker image with 600+ pulls in docker-hub

Technical Skills

Languages: Python, C++, SQL, HTML, CSS
Libraries: PyTorch, FastAPI, Scikit-learn, TensorFlow (Keras), NumPy, Pandas, Flask
Tools: Docker, Git, GitHub, Linux, Ollama, Postman

Publications

HybridStack-MLP: Advanced Ensemble Learning for Malicious QR Code and URL Detection — 2025
- Presented at *2nd IEEE International Conference for Women in Computing (InCoWoCo 2025)*; to appear in IEEE Xplore.
- Ensemble learning model achieving 91.3% accuracy in malicious QR code and phishing URL detection.

Achievements

Winner — State Level Smart Hackathon (SLSH), Himachal Pradesh — 2025